

9.2.2

EE25BTECH11042 - Nipun Dasari

Question:

Find the points of intersection of the following parabola and circle and plot the region

$$y^2 \leq 6ax \quad (0.1)$$

$$x^2 + y^2 \leq 16a^2 \quad (0.2)$$

Solution:

We first express the equations for the parabola (0.1) and the circle (0.2) in the standard quadratic form $g(x, y) = 0$.

$$y^2 - 6ax = 0 \quad (0.3)$$

$$x^2 + y^2 - 16a^2 = 0 \quad (0.4)$$

In homogeneous coordinates, using the form $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$, where $\mathbf{x} = \begin{pmatrix} x & y & 1 \end{pmatrix}^T$, the matrices for the conics are:
where,

$$\mathbf{C} = \begin{pmatrix} \mathbf{V} & \mathbf{u} \\ \mathbf{u}^T & f \end{pmatrix} \quad (0.5)$$

$$\mathbf{C}_1 = \begin{pmatrix} 0 & 0 & -3a \\ 0 & 1 & 0 \\ -3a & 0 & 0 \end{pmatrix} \quad (0.6)$$

$$\mathbf{C}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -16a^2 \end{pmatrix} \quad (0.7)$$

The intersection points lie on a degenerate conic formed from a linear combination $P_1 + \mu C_2 = 0$. We must find the value of μ that makes the determinant of the combined conic's matrix, $C(\mu) = C_1 + \mu C_2$, equal to zero. The general matrix for this combination is:

$$C(\mu) = \begin{pmatrix} \mu & 0 & -3a \\ 0 & 1 + \mu & 0 \\ -3a & 0 & -16a^2\mu \end{pmatrix} \quad (0.8)$$

We set the determinant to zero. By applying the row operation $R_3 \rightarrow R_3 + \frac{3a}{\mu} R_1$, which

does not change the determinant's value, we get an upper triangular matrix:

$$\begin{aligned}
 \det(C(\mu)) &= \det \begin{pmatrix} \mu & 0 & -3a \\ 0 & 1 + \mu & 0 \\ 0 & 0 & -16a^2\mu - \frac{9a^2}{\mu} \end{pmatrix} \\
 &= (\mu)(1 + \mu) \left(-16a^2\mu - \frac{9a^2}{\mu} \right) \\
 &= (1 + \mu)(-16a^2\mu^2 - 9a^2) \\
 &= -a^2(1 + \mu)(16\mu^2 + 9) = 0
 \end{aligned} \tag{0.9}$$

The only real root is obtained from the first factor, which gives $\mu = -1$. The equation for this degenerate conic is found by subtracting (0.4) from (0.3):

We use the general conic (quadratic) matrix representation:

$$\begin{pmatrix} x & y \end{pmatrix} Q \begin{pmatrix} x \\ y \end{pmatrix} + L^\top \begin{pmatrix} x \\ y \end{pmatrix} + C = 0, \tag{0.10}$$

where

$$Q = \begin{pmatrix} A & H/2 \\ H/2 & B \end{pmatrix}, \quad L = \begin{pmatrix} G \\ F \end{pmatrix}. \tag{0.11}$$

Simplify:

$$-x^2 - 6ax + 16a^2 = 0. \tag{0.12}$$

Hence,

$$Q = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} -6a \\ 0 \end{pmatrix}, \quad C = 16a^2. \tag{0.13}$$

Matrix form:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -6a & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 16a^2 = 0. \tag{0.14}$$

Same as above:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -6a & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 16a^2 = 0. \tag{0.15}$$

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 6a \\ 0 \end{pmatrix}, \quad C = -16a^2. \tag{0.16}$$

Matrix form:

$$\begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 6a & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - 16a^2 = 0. \tag{0.17}$$

This represents two linear equations:

$$x + 8a = 0 \quad \text{or} \quad x - 2a = 0. \tag{0.18}$$

Matrix (vector) form of each line:

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + 8a = 0 \quad \text{and} \quad \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - 2a = 0. \quad (0.19)$$

This degenerate conic consists of two vertical lines, $x = 2a$ and $x = -8a$. The line $x = -8a$ does not intersect the parabola $y^2 = 6ax$ in real points (assuming $a > 0$). We thus consider the line $x = 2a$.

We find the intersection of the line $x - 2a = 0$ with the parabola P_1 using the formula $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$. The line and conic parameters from P_1 are: $\mathbf{h} = \begin{pmatrix} 2a \\ 0 \end{pmatrix}$, $\mathbf{m} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $\mathbf{V} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} -3a \\ 0 \end{pmatrix}$, and $f = 0$.

Next, we compute the components for the parameter κ_i :

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = 1 \quad (0.20)$$

$$g(\mathbf{h}) = -12a^2 \quad (0.21)$$

$$\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) = 0 \quad (0.22)$$

We substitute these into the formula for κ_i :

$$\begin{aligned} \kappa_i &= \frac{1}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{V} \mathbf{m})} \right) \\ &= \frac{1}{1} \left(-(0) \pm \sqrt{(0)^2 - (-12a^2)(1)} \right) = \pm \sqrt{12a^2} = \pm 2\sqrt{3}a \end{aligned} \quad (0.23)$$

This gives two values: $\kappa_1 = 2\sqrt{3}a$ and $\kappa_2 = -2\sqrt{3}a$.

Finally, we find the intersection points using $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$:

$$\mathbf{x}_1 = \begin{pmatrix} 2a \\ 0 \end{pmatrix} + 2\sqrt{3}a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a \\ 2\sqrt{3}a \end{pmatrix} \quad (0.24)$$

$$\mathbf{x}_2 = \begin{pmatrix} 2a \\ 0 \end{pmatrix} - 2\sqrt{3}a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2a \\ -2\sqrt{3}a \end{pmatrix} \quad (0.25)$$

Thus, the intersection points are $(2a, 2\sqrt{3}a)$ from (0.24) and $(2a, -2\sqrt{3}a)$ from (0.25).

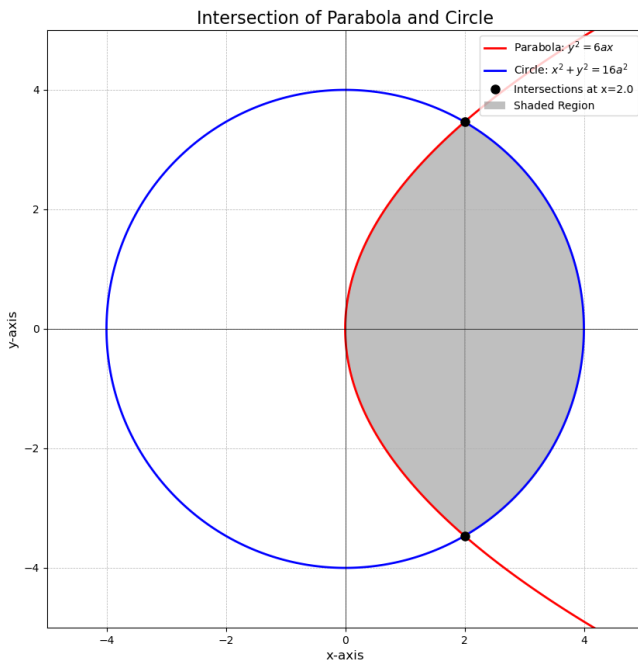


Fig : Parabola and Circle

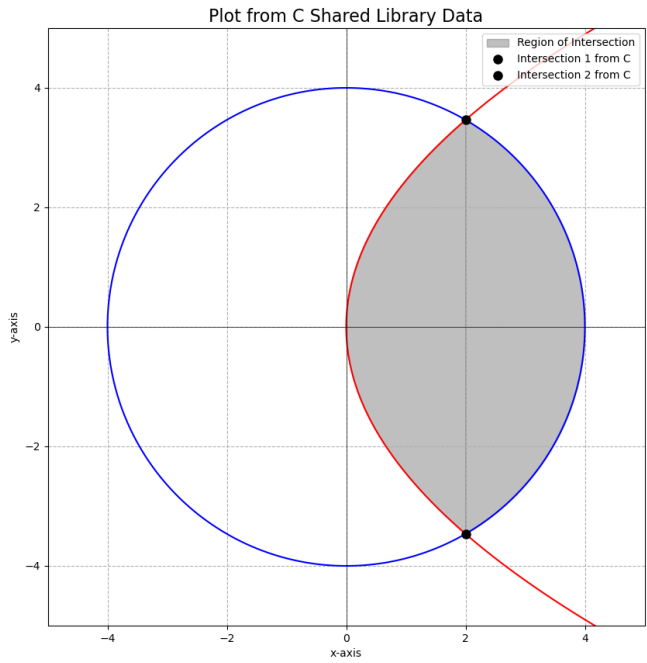


Fig : Intersection Points